

E-Commerce Sales Analysis & Budget Tracking System

Project Protocol / Documentation

1. Project Overview

This project analyzes e-commerce sales data, tracks revenue, monitors budgets, provides business insights, and supports secure access using JWT authentication.

2. Technologies Used

Frontend: HTML, CSS, Bootstrap, React

Backend: Java, Spring Boot

Database: MySQL

Analytics: Power BI, Charts

Security: JWT Authentication, BCrypt

3. Main Goals

- Analyze sales performance
- Track revenue and expenses
- Monitor budget utilization
- Identify top-selling products
- Analyze customer behavior
- Generate business reports

4. System Workflow

User Login → JWT Authentication → Product Management → Order Processing → Sales Analysis → Budget Analytics → Dashboard & Reports

5. Algorithms Used

1. Sales Trend Analysis
2. Budget Utilization Analysis
3. Top Selling Product Algorithm
4. Customer Purchase Frequency Analysis
5. Revenue Calculation Algorithm
6. Collaborative Filtering
7. Linear Regression Forecasting
8. JWT + BCrypt Security

6. JWT Authentication Process

User logs in → Credentials verified → JWT token generated → Token stored in frontend → Token sent with API requests → Backend validates token → Access granted.

7. Advantages

- Secure authentication
- Fast access control
- Better sales monitoring
- Improved business decisions
- Easy data visualization

8. Viva Questions & Answers

Q: Why JWT?

A: To secure APIs and control user access.

Q: Why Power BI?

A: To create interactive dashboards and reports.

Q: What is the main objective?

A: To analyze e-commerce sales and support data-driven decisions.